

Matteo Merler

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Summary

I am a researcher at FBK NLP in Trento, Italy, and I work with Tom Silver's group in Princeton on robotics and planning. This Fall, I will start my PhD at the Bethge Lab in Tübingen as an ELLIS and IMPRS-IS PhD student.

I want to build AI agents that learn from experience and adapt quickly to new situations. My recent work uses language models to write executable world models and policies: programs that can be tested, revised, and reused to plan and act, in both simulated games and real-world robotics.

My work has been published at NeurIPS (2024, 2026) and EMNLP 2026. My MSc thesis at Aalto University, published at NeurIPS 2024, received the School of Science Master's Thesis Award, given to three theses among 517 MSc graduates in 2024. I graduated with honors from both my MSc and my BSc at the University of Trento.

Selected Publications

Coding Agents for Generalized Task and Motion Planning Problems Sept 2026
Matteo Merler, Bowen Li, Josh Roy, Yichao Liang, Qianwei Wang, Yixuan Huang, Tom Silver
arXiv preprint

Generating Code World Models with Large Language Models Guided by Monte Carlo Tree Search Dec 2024
Nicola Dainese†, **Matteo Merler**†, Minttu Alakuijala, Pekka Marttinen
Advances in Neural Information Processing Systems 37 (NeurIPS 2024)

† *denotes equal contribution*

Education

M.Sc. **Aalto University**, Machine Learning, Data Science and Artificial Intelligence — Sept 2022 – Sept 2024

- Graduated with honors, with a **final grade of 5.0/5.0** and a **GPA of 5.0/5.0**.
- Thesis: **Generating Code World Models with Large Language Models Guided by Monte Carlo Tree Search**, supervised by Prof. Pekka Marttinen, published at NeurIPS 2024.
- Thesis received the School of Science **Master's Thesis Award**, given to three theses among 517 MSc graduates in 2024.
- Received the Dean's Scholarship for outstanding academic performance.

B.Sc. **University of Trento**, Computer Science — Sept 2019 – July 2022

- Graduated with a **final grade of 110/110 cum laude (with honors)** and a **GPA of 28.46/30**.
- Thesis: **Route Optimization for Indoor Spaces**, supervised by Prof. Alberto Montresor (in Italian).

H.S. **Liceo Scientifico Leonardo da Vinci**, Liceo Scientifico (Italian Scientific High School Diploma) — Sept 2014 – July 2019

- Graduated with a **final grade of 100/100**.

Experience

Researcher, Fondazione Bruno Kessler — Trento, Italy
Working as a pre-doctoral researcher ("Researcher 4 contract") in the Natural Language Processing Lab under the supervision of Bernardo Magnini. — Jan 2025 – present

- Lead researcher on FBK's side of a joint project on **planning with Vision-Language Models** (with Aalto University), under resubmission.
- Led a second project on **selectively integrating VLM guidance with model-free Reinforcement Learning** based on the agent's own uncertainty, with a preliminary version accepted at the CAIPI workshop at ECAI 2025 and a full paper in preparation.
- Participating on private research for collaborations with companies.
- Contributing to the draft of a **Horizon Europe funding proposal**, leading the lab's team for Language and Reinforcement Learning.
- Active participation to the research group's scientific activities, including monthly meetings and collaborative discussions.

Research Assistant, Aalto University — Helsinki, Finland
MSc Thesis worker in the Foundation Models for Language and Reinforcement Learning (Larel) group, supervised by Prof. Pekka Marttinen. — Aug 2024 – Sept 2024

- Conducted a research project on **Symbolic Regression with Large Language Models**, presented at the ACL Student Research Workshop.
- Co-led a research project on **Code World Models** with PhD candidate Nicola Dainese, forming the basis of my Master's Thesis, accepted at NeurIPS 2024.
- Started a project on **planning with Vision-Language Models**, later continued by both Aalto University and my current group at FBK.
- Regularly attended lab meetings and participated in broader academic activities, including the Seminar on Large Language Models.

Internship, ThinkIN — Trento, Italy
Worked as a software developer intern in collaboration with the University of Trento. — Feb 2022 – June 2022

- Designed and built a prototype for an indoor navigation system specifically targeted for supermarkets, to be used with the company's own location tracking system.

Publications

QVal: Cheaply Evaluating Dense Supervision Signals for Long-Horizon LLM Agents — Dec 2026
Sergio Hernández-Gutiérrez, **Matteo Merler**†, Ilze Amanda Auzina†, Joschka Strüber, Ameya Prabhu, Matthias Bethge
Advances in Neural Information Processing Systems 39 (NeurIPS 2026), Evaluations and Datasets Track

Selective Agent Guidance via Entropy: Learning Autonomous Policies from Imperfect VLM Teachers — Oct 2026
Giovanni Bonetta†, **Matteo Merler**†, Davide Zago, Rossella Cancelliere, Bernardo Magnini
Findings of the Association for Computational Linguistics: EMNLP 2026

DecSelfMask: Leveraging Unlabeled Text via Self-Relevance-Guided Masking for Decoder-Only Classification — Oct 2026
Pietro Ferrazzi, **Matteo Merler**, Giovanni Bonetta, Alberto Lavelli, Bernardo Magnini
Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)

ViPlan: A Benchmark for Visual Planning with Symbolic Predicates and Vision-Language Models Oct 2026
Matteo Merler†, Nicola Dainese†, Minttu Alakuijala, Giovanni Bonetta, Pietro Ferrazzi, Yu Tian, Bernardo Magnini, Pekka Marttinen
Findings of the Association for Computational Linguistics: EMNLP 2026

EMPIRIC: Experiment-Driven Learning of Residual World Models for Robot Planning Sept 2026
Yichao Liang, Amber Li, Dat Nguyen†, Emily Bunnapradist†, Michelangelo Naim†, Sreela Kodali†, **Matteo Merler**, Bowen Li, Kiran Gopinathan, Yiyun Liu, Nikhil Pimpalkhare, Joshua B. Tenenbaum, Adrian Weller, Zenna Tavares, Tom Silver, Kevin Ellis
arXiv preprint

Coding Agents for Generalized Task and Motion Planning Problems Sept 2026
Matteo Merler, Bowen Li, Josh Roy, Yichao Liang, Qianwei Wang, Yixuan Huang, Tom Silver
arXiv preprint

Guiding Reinforcement Learning with Selective Vision-Language Model Supervision Oct 2025
Matteo Merler, Giovanni Bonetta, Bernardo Magnini
ECAI Workshop on AI-based Planning for Complex Real-World Applications (CAIPI'25)

Generating Code World Models with Large Language Models Guided by Monte Carlo Tree Search Dec 2024
Nicola Dainese†, **Matteo Merler**†, Minttu Alakuijala, Pekka Marttinen
Advances in Neural Information Processing Systems 37 (NeurIPS 2024)

In-Context Symbolic Regression: Leveraging Large Language Models for Function Discovery Aug 2024
Matteo Merler†, Katsiaryna Haitsiukevich†, Nicola Dainese†, Pekka Marttinen
Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics, Volume 4: Student Research Workshop (ACL 2024 SRW)

† *denotes equal contribution*

Additional Experience and Awards

Master's Thesis Award: My Master's Thesis received the School of Science Master's Thesis Award, given to three theses among 517 MSc graduates in 2024.

Volunteer

Volunteer, Libera Associazioni Nomi e Numeri Contro le Mafie — Calabria, Italy
July 2017 – July 2017
I participated in a volunteering camp experience in southern Italy where we cultivated the fields of mafia-confiscated houses for the Libera association, while also taking part in various activities on the territory and listening to testimonies from victims and policemen.

Volunteer, Associazione Volontaria "Per un Mondo Migliore" — Rijeka, Croatia
Aug 2018 – Aug 2018
I participated in a volunteering camp experience in Rijeka, Croatia where we helped different families and associations in need around the city and in mainland Croatia while living together in a shared home.

Languages

Italian: Native | **English:** C2 (Near-native proficiency) | **German:** A2 (Basic)